Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Simulate smoothing and sharpening operations on images using frequency domain filters.**	
Experiment No: 10	**Date:29-10-2026**	**Enrollment No:92400133121**

Aim: Simulate smoothing and sharpening operations on images using frequency domain filters.

Theory:
Smoothing and sharpening operations can also be performed in the frequency domain using filters such as the Gaussian filter and the Laplacian filter.

Smoothing in Frequency Domain:
To perform smoothing in the frequency domain, we apply a low-pass filter that attenuates high-frequency components. This helps to blur the image and reduce noise. One common approach is to use a Gaussian filter in the frequency domain.

Sharpening in Frequency Domain:
To perform sharpening in the frequency domain, we apply a high-pass filter that enhances high-frequency components. This amplifies the edges and fine details in the image. One common approach is to use a combination of a high-pass filter and the original image.

Code:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Read image using image path
img = cv2.imread('images/image_1.jpg', 0)

# 2. Fourier Transform
F = np.fft.fftshift(np.fft.fft2(img))
M, N = img.shape

# 3. Create frequency distance matrix
u, v = np.meshgrid(np.arange(N)-N//2, np.arange(M)-M//2)
D = np.sqrt(u**2 + v**2)

# 4. Low-Pass and High-Pass Filters
LPF = D < 30
HPF = D >= 30

# 5. Apply filters
smooth = np.abs(np.fft.ifft2(np.fft.ifftshift(F * LPF)))
high_pass = np.abs(np.fft.ifft2(np.fft.ifftshift(F * HPF)))
```

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```
sharp = np.clip(img.astype(float) + high_pass, 0, 255)

# 6. Display images
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')

plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothening')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')
plt.show()
```

Output:



Conclusion:

→Smoothing: The image was blurred by removing high-frequency components.
→Sharpening: The edges and fine details of the image were enhanced using a high-pass filter.
→The Fourier transform was used to process the image in the frequency domain.
→The original, smoothed, and sharpened images were displayed and compared.

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